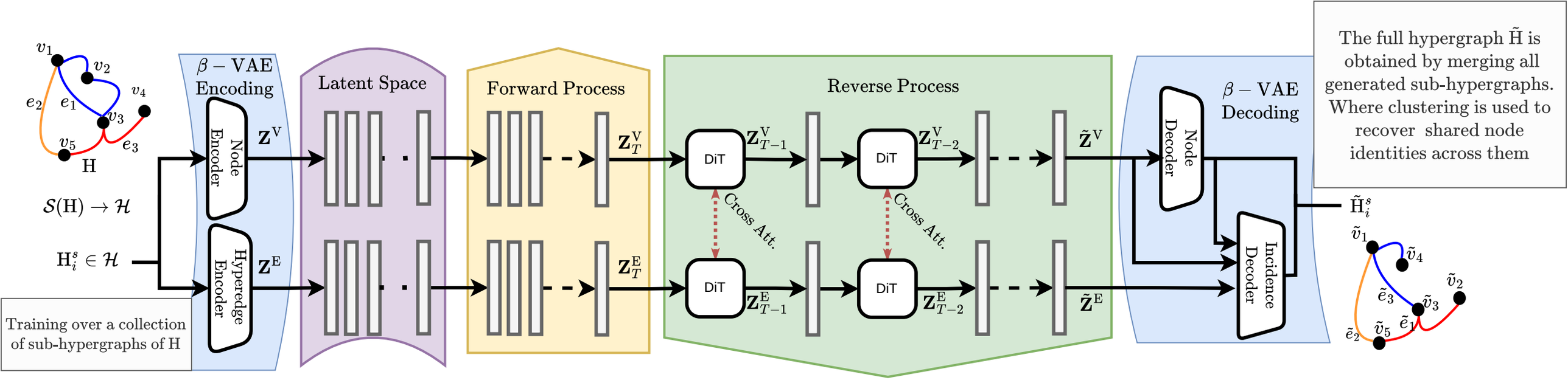

v_1
v_2
v_3
v_4
v_5
e_1
e_2
e_3
H
$\mathcal{S}(\mathrm{H}) \to \mathcal{H}$
$\mathrm{H}_i^s \in \mathcal{H}$
Training over a collection of sub-hypergraphs of H
$\beta-$ VAE Encoding
Node Encoder
Hyperedge Encoder
$\mathbf{Z}^{\mathrm{V}}$
$\mathbf{Z}^{\mathrm{E}}$
Latent Space
Forward Process
$\mathbf{Z}_T^{\mathrm{V}}$
$\mathbf{Z}_T^{\mathrm{E}}$
Reverse Process
DiT
$\mathbf{Z}_{T-1}^{\mathrm{V}}$
$\mathbf{Z}_{T-2}^{\mathrm{V}}$
$\tilde{\mathbf{Z}}^{\mathrm{V}}$
Cross Att.
$\mathbf{Z}_{T-1}^{\mathrm{E}}$
$\mathbf{Z}_{T-2}^{\mathrm{E}}$
$\tilde{\mathbf{Z}}^{\mathrm{E}}$
$\beta-$ VAE Decoding
Node Decoder
Incidence Decoder
$\tilde{\mathrm{H}}_i^s$
$\tilde{v}_1$
$\tilde{v}_2$
$\tilde{v}_3$
$\tilde{v}_4$
$\tilde{v}_5$
$\tilde{e}_1$
$\tilde{e}_2$
$\tilde{e}_3$
The full hypergraph $\tilde{\mathrm{H}}$ is obtained by merging all generated sub-hypergraphs. Where clustering is used to recover shared node identities across them